
HIC 2026
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Innovating water solutions for climate changes
June 22-26, 2026, Zaragoza, Spain
TRACKING HYDROLOGICAL DROUGHT PROPAGATION IN RIVER NETWORKS: AN EVENT-BASED FRAMEWORK
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TRACKING HYDROLOGICAL DROUGHT PROPAGATION IN RIVER NETWORKS: AN EVENT-BASED FRAMEWORK

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1. MOTIVATION / PROBLEM

- Hydrological droughts are dynamic phenomena that propagate in space and time
- Not a simple linear process Integrative systems (upstream conditions affecting downstream reaches)
- Approaches:
  - **Clustering algorithms** (Janzing et al., (2026)), **individual event perspective** (Laaha et al., (2017)), **total area under drought conditions** (Hanel et al., 2018; Wanders et al., 2015), **hydrological drivers** (Colombo et al., 2023; Van Tiel et al., 2023), **multiple basins** (Teutschbein et al., 2022; Brunner et al., 2021; Barker et al., 2019)
  - Gap: Explicit analysis of hydrological drought propagation from an **event-based perspective**
- **Objective:** Track and quantify hydrological propagation dynamics using an even-based framework

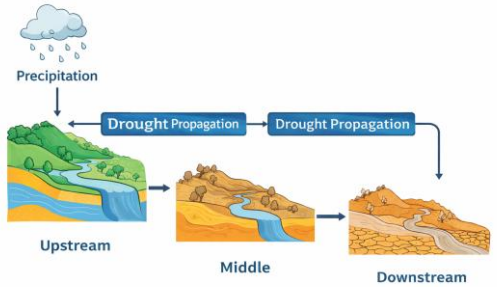


Figure 1. Hydrological drought propagation simple scheme

2. METHOD / ALGORITHM

- Drought event identification (SSI, Vicente Serrano et al., (2025))
- Network connectivity
- Link upstream – downstream events **(search window (W) and overlap filter (M))**

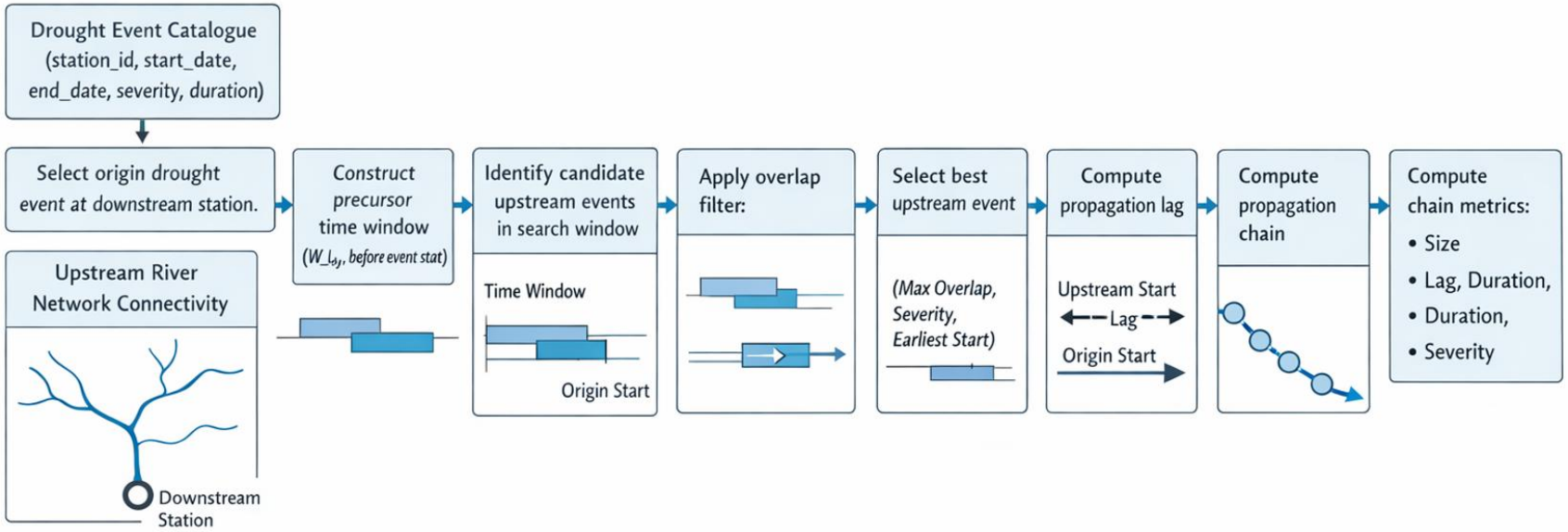


Figure 2. Conceptual workflow of the hydrological drought propagation algorithm used to identify upstream–downstream drought relationships within the river network. Starting from a drought event catalogue and the upstream connectivity structure of the river network, the method selects a downstream origin event and constructs a precursor search window extending W days before the event onset. Candidate drought events at upstream stations intersecting this window are identified and filtered according to a minimum temporal overlap criterion. When multiple candidates exist at a given station, the best upstream event is selected using deterministic tie-breaking rules.

3. CHAIN METRICS

Descriptor	Symbol	Description	Expression	Units
Chain start	$t_{(c,s)}$	Earliest onset among all chain events	$t_{(c,s)} = \min(t_{s,k}), k \in C$	—
Chain end	$t_{(c,e)}$	Latest termination among all chain events	$t_{(c,e)} = \max(t_{e,k}), k \in C$	—
Chain duration	D_c	Total span, from first onset to last termination	$D_c = t_{(c,e)} - t_{(c,s)} + 1$	days
Chain size	N_c	Total stations in chain (upstream matched + origin)	$N_c = N_{up} + 1$	—
Propagation fraction	f_p	Fraction of upstream network affected	$f_p = N_{up} / N_{avail}$	—
Mean lag	λ	Mean onset lag between upstream events and origin event	$\lambda = (N_{up})^{-1} \Sigma \lambda_k, k \in C$	days
Chain severity	$S_{(c)}$	Cumulative SSI-based severity across all chain events	$S_c = \Sigma S_k, k \in C$	—
Chain volumetric severity	$V_{(c)}$	Total discharge deficit volume across all chain events	$V_c = \Sigma V_k, k \in C$	hm^3

C = set of events forming the propagation chain; k indexes individual events within the chain
N_{up} = total number of upstream stations connected to the origin station.
N_{avail} = upstream stations present in event catalogue

Table 1. Mathematical definition of the propagation chain descriptors. Chain properties include start and end dates, duration, propagation size, propagation fraction, mean lag, chain severity and chain severity in volume (hm3) units.

4. STUDY AREA / DATA

- *Centro de Estudios y Experimentación de Obras Públicas* (CEDEX, https://www.cedex.es). Period 1961-2020

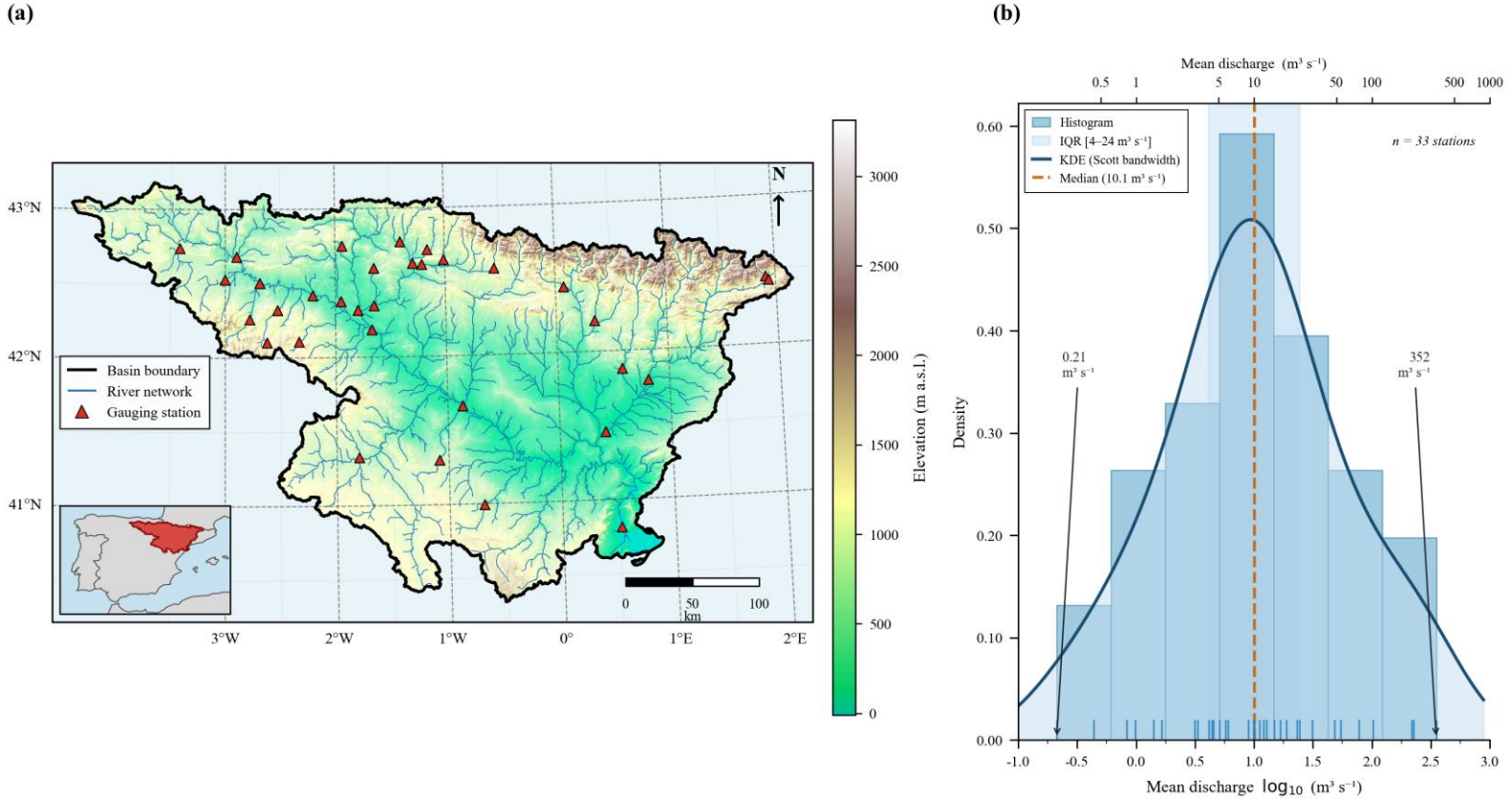


Figure 3. (a) Study area showing the Ebro River Basin (Spain), including the river network and the location of the 33 gauging stations used in this study. Elevation is represented as a shaded background. (b) Distribution of mean discharge across the gauging stations, expressed in $\log_{10}$ scale. The histogram and kernel density estimate illustrate the wide range of flow conditions represented in the dataset, from low-discharge headwater stations to high-flow downstream sections.

5. RESULTS - PROPAGATION

- 525 propagation chains (**23%** of the 2265 original drought events)
- Duration: months to years
- Chain size / Propagation fraction: From few stations implicated to great part of the network
- Lag: days to months
- Severity: great variability, ranging from little impact on the hydrological system to events with a large impact

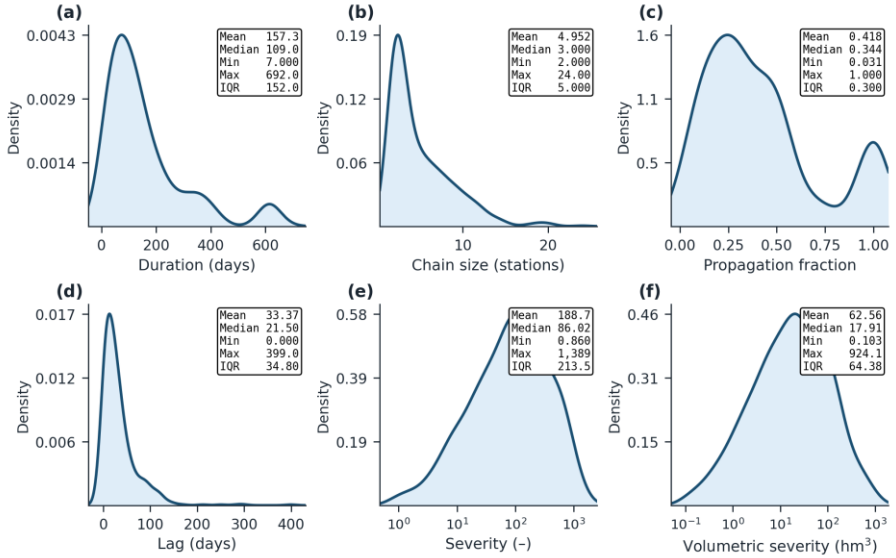


Figure 4. Distribution of hydrological drought propagation chain characteristics: (a) chain duration, (b) chain size, (c) propagation fraction, (d) propagation lag, (e) chain severity, and (f) volumetric severity. Shaded areas represent kernel density estimates.

6.a CASE STUDIES – BASIN WIDE PROPAGATION EVENT

Propagation chain 9027_38 | Origin: Station 9027 | 21 Sep 1988 – 24 Apr 1989

D = 216 days | N = 24 stations | fp = 0.719 | Mean lag = -35.8 d | Severity = 1160.2 | Vol. severity = 896.6 hm³

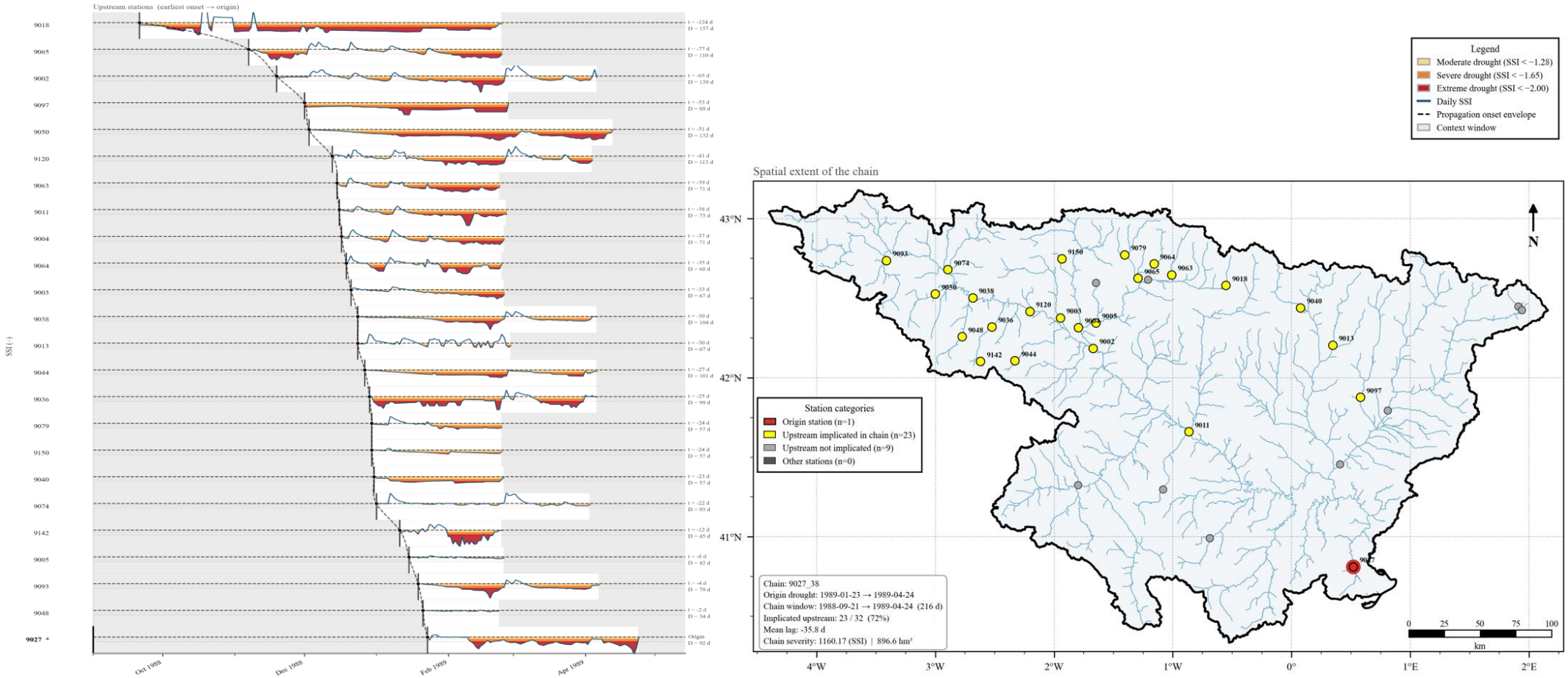


Figure 5. Spatiotemporal representation of the hydrological drought propagation chain associated with the event originating at station 9027 (Ebro River at Tortosa).

6.b CASE STUDIES – LOCALIZED PROPAGATION EVENT

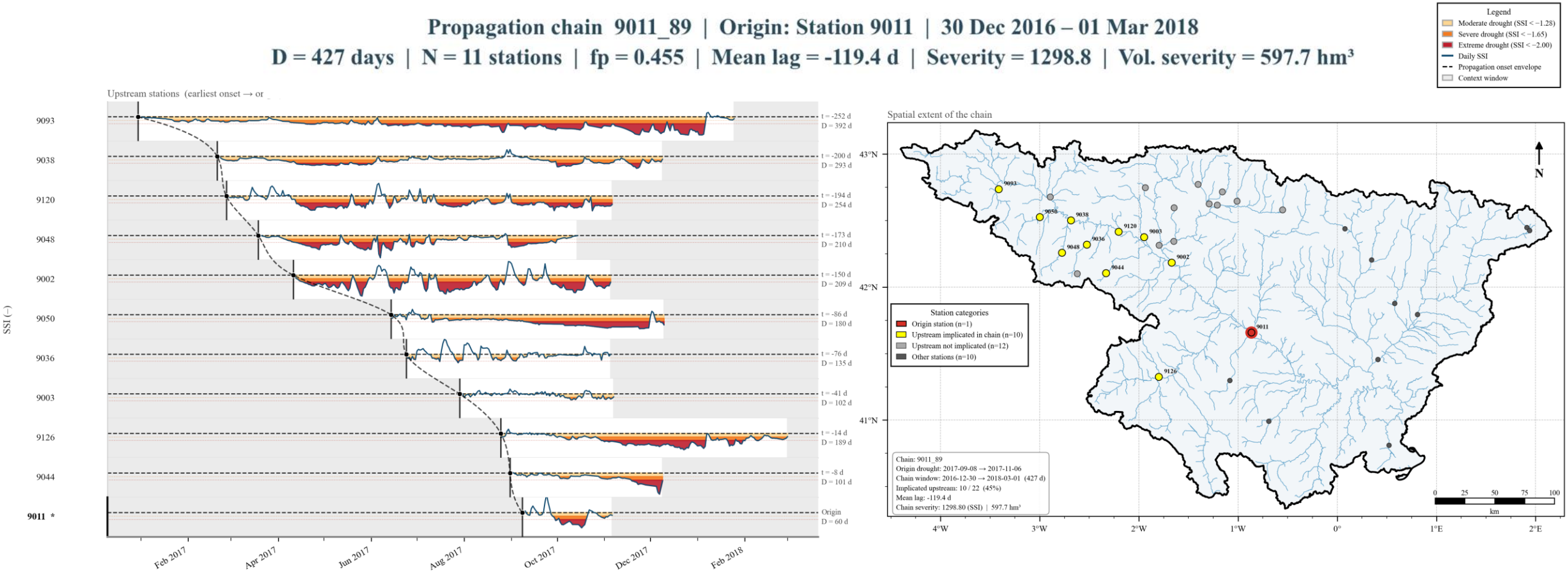


Figure 6. Spatiotemporal representation of the hydrological drought propagation chain associated with the event originating at station 9011 (Ebro River at Zaragoza).

6.c CASE STUDIES – RESERVOIR DYNAMICS INDUCED EVENT

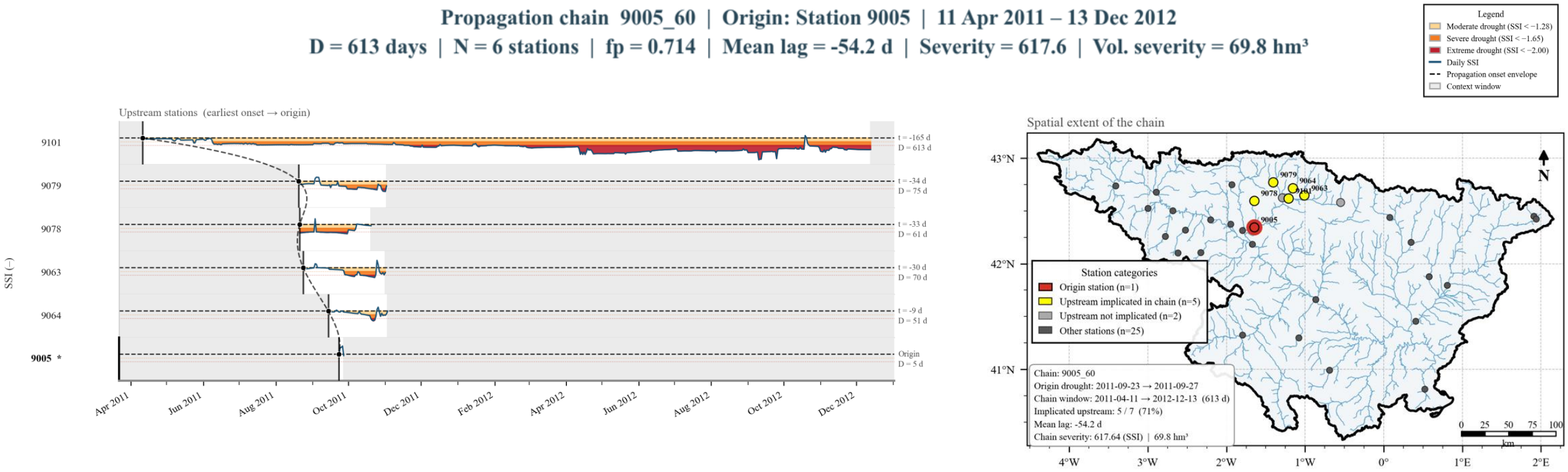


Figure 7. Spatiotemporal representation of the hydrological drought propagation chain associated with the event originating at station 9005 (Aragón River at Caparroso).

7. FUTURE RESEARCH DIRECTIONS

- Multi-basin extension across Western Europe (Spain, France, UK, Ireland, Germany, Italy): comparative analysis of propagation dynamics across contrasting climate regimes and regulation intensities (1980–2020)
- Integration of explicit reservoir metrics and comparison (regulated vs non regulated basins)
- Upward extension to meteorological drought: reconstructing full SPI→SSI propagation pathways within a network-aware, event-based framework
- PIML / PIDL / DL for drought forecasting: embedding spatial coherence and water balance constraints to anticipate downstream drought from upstream signals

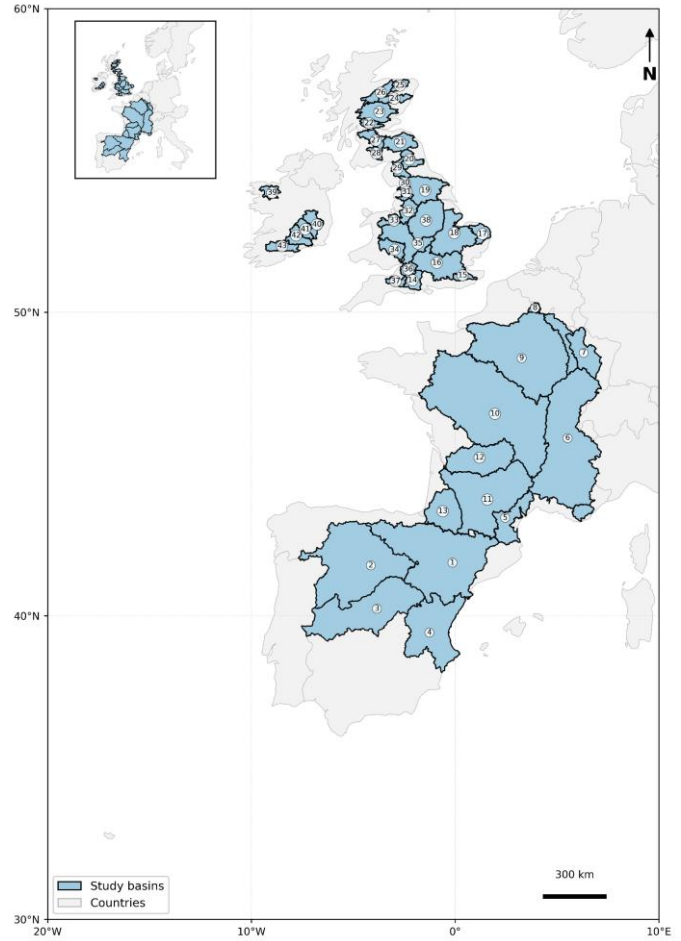


Figure 8. Study area showing the river basins selected across Western Europe. Blue polygons indicate the river basins included in the multi-basin propagation analysis. Inset map shows the full spatial extent of the study domain.

8. TAKE-HOME MESSAGES

- **Method**
  - New scalable **event-based framework** to track hydrological drought propagation in river networks
  - Reconstructs **upstream-downstream propagation chains**
- **Findings**
  - Hydrological drought propagation is **highly variable across space and time**
  - Ranges from local sub-basin events to basin-wide processes
- **Impact**
  - Provides a **process-based understanding** of hydrological droughts
  - Useful for water management and early warning systems

"Hydrological droughts are not isolated events – they propagate through different reaches of the river network"

THANK YOU FOR YOUR ATTENTION!





